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SAN FRANCISCO STATE UNIVERSITY

Graduate Studies ADM-250 (415) 338-2234

Proposal for Culminating Experience (PCE) APPROVAL FORM

Name: Naisarg Dharmendra Halvadiya Student ID: 923613129

SFSU Email: nhalvadiya@sfsu.edu Degree and Major: Data Science and Artificial Intelligence

Concentration or Emphasis (if applicable): _____

Attach Completed PCE form here:	If Applicable, Attach IRB Approved Protocol or ADE Notification Here:	If Applicable, Attach Non-SFSU Faculty CV Here:

FOR COMMITTEE CHAIR:

I will be available to work with students (1) during winter break ☐ YES ☒ NO (2) during the summer months ☒ YES ☐ NO

Name	Signature	Department	SFSU Email	Date
Shahrukh Humayoun	DocuSigned by: Shahrukh Humayoun 94E06432EAE246C	Computer Science	humayoun@sfsu.edu	03/26/2024 | 2:39 PM PDT

Second Committee Member (Required):

Name	Signature	Department	Email	Date
Jingyi Wang	DocuSigned by: Jingyi Wang 7CFEB96626BA4CF	computer Science	jingyiwang@sfsu.edu	03/26/2024 | 9:09 PM PDT

Third Committee Member (as necessary):

Name	Signature	Department	Email	Date

Department Chair/Graduate Coordinator:

I have reviewed the proposal, including the composition of the supervising committee, and find it acceptable for meeting the culminating experience requirement for the master's degree in the major indicated.

Name	Signature	Department	SFSU Email	Date
Dr. Hui Yang	DocuSigned by: 7512C57EEF6A46F	computer science	huiyang@sfsu.edu	04/01/2024 | 6:06 PM PDT

FOR GRADUATE STUDIES OFFICE USE ONLY

ORSP determination: Approved protocol # _____ Exempt #_____ Not Human/Animal Subject or Research ☐

Date of ORSP determination (if known): _____ Pending HAP Review/ Approval ☐

Approved ☒ Not Approved ☐
Pending ORSP Determination ☐

DocuSigned by: 3D68EB20C83143E... Dean of the Graduate Division or Designee — Date: 04/03/2024 | 10:31 AM PDT

Note: Upon approval of the PCE, read graduate Academic Policies and Procedures section in the Bulletin regarding conditions for maintaining its validity.

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Division of Graduate Studies
1600 Holloway Avenue, ADM 250
San Francisco, CA 94132
Phone: 415/338-2234
gradstdy@sfsu.edu
web: grad.sfsu.edu

Culminating Experience Procedures

Instructions for Completing Proposal for Culminating Experience (PCE)

The culminating experience must be met by the satisfactory completion of a thesis, special project, comprehensive examination, and/or an oral defense of the work (courses: 890, 892, 893, 894, 895, 896EXM, 898, 998).

Steps to Completion:

Submit your Proposal for Culminating Experience (PCE) through DocuSign by the deadline a semester prior to registering in your culminating experience course. **The deadlines for submission to Graduate Studies are April 1 for fall course enrollment and November 1 for spring course enrollment.**

Your PCE must be approved by your Culminating Experience Committee and the Department Chair or Graduate Coordinator for your program.

Selection of Culminating Experience Committee:

The committee must consist of a minimum of two members. Some programs require a third member. The chair and the second member of your committee must hold tenured/tenure-track faculty appointments in your major department.

In circumstances where special expertise is available in another department, the graduate dean may authorize a designated tenured/tenure-track faculty member from a related department to serve as second reader.

The third member of a committee may be a lecturer or from outside the major department or university. Lectures or readers outside the university must have a current curriculum vitae on file in the Division of Graduate Studies.

With special permission from the Dean of Graduate Studies, some long-term lecturers with terminal degrees in their field or with special expertise may serve as the second reviewer provided their curriculum vitae is on file in the Division of Graduate Studies.

To officially change the composition of your committee, you must submit a Petition for Committee Revision to culminatingexperience@sfsu.edu.

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Research Involving Humans, Vertebrate Animals or Biological Specimens:

If your project includes research with humans, vertebrate animals, or biological specimens (e.g., tissues, stem cells/cell lines, blood), additional documentation is required. Refer to Section 8 of the PCE form.

If you are unsure if your project involves human subjects visit the website of the Office of Research and Sponsored Programs - Human and Animal Protections (ORSP-HAP) at http://research.sfsu.edu/protocol/.

IMPORTANT: You may NOT begin your research until you obtain an official notice of Approval, Exception or Exemption from ORSP-HAP

Registration and Grading for Culminating Experience Courses:

Registration in a Culminating Experience course will not be permitted until your Advancement to Candidacy (ATC) and Proposal for Culminating Experience (PCE) are approved by the Division of Graduate Studies. You must have a 3.0 GPA in all post-baccalaureate course work completed.

Important Notes:

If you do not complete the course by the end of the semester of registration and receive a grade of RP (Report in Progress), do not register for the course again. When the project is completed, be sure that your committee chair files a grade change to CR (Credit) with the Registrar's Office.

A Report of Completion must be filed for internships, field studies, or creative work projects. A Thesis Receipt signed by Graduate Studies is required for thesis or written creative work.

Students must adhere to the Continuous Enrollment Policy as stated in the bulletin. After the semester of enrollment in the Culminating Experience and the subsequent (grace period) semester, all graduate students are required to maintain continuous enrollment through the College of Extended Learning (CEL) or a degree-related course in their field while completing the culminating experience.

Form begins on Page 3

PLEASE NOTE: in order save your personal information on the following PDF forms, you will need to:

1. Install latest version of Adobe Acrobat Reader on your computer. Click here for Free Adobe Acrobat Readerhttp://www.adobe.com/products/acrobat/readstep2.html.
2. Save the PDF form to your computer desktop prior to entering your personal information.

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PROPOSAL FOR CULMINATING EXPERIENCE

895: FIELD STUDY OR APPLIED RESEARCH

REPORT OF COMPLETION REQUIRED

Complete, print and file this form with the Division of Graduate Studies in accordance with guidelines published in the University *Bulletin*.

NO HANDWRITTEN FORMS WILL BE ACCEPTED

1. Official Degree Title as listed in the University Bulletin:

Master of Science Major Data Science & Artificial Intelligence

Concentration or emphasis (if applicable) _____

2. Name Naisarg Dharmendra Halvadiya Student ID 923613129

Address 100 Font Blvd Phone 6289460347

City/State/Zip San Francisco, CA, 94132 Email nhalvadiya@sfsu.edu

3. ☐ **Check here if this is a REVISED proposal (withdrawing previous proposal)**

4. ☑ **I plan to register for the 895 course in (enter term and year):** Fall 2024 **or** ☐ **I previously registered for the 895**

5. Title (*Limit 12 words*): (Report any title change to the Division of Graduate Studies prior to filing completed work.)

A Visual Analytics tool to Compare People's Emotions and Feelings in Tweets

6. Brief statement of project and research methods (*must fit in space allotted*):

Twitter (X) is one of the popular social media platforms where people share news or reactions towards an event or topic using short text messages called "tweets" (now posts). Emotion analysis in these tweets can play a vital role in understanding peoples' feelings towards the underlying event or topic. In this project, I will develop a visual analytics tool to provide comparison views of peoples' emotional feelings in tweets towards an event or topic. Furthermore, I'll provides several interactions and filtering options for navigation and better exploration of underlying tweet data for emotional feelings' comparison.

7. Projected timeline for completion of culminating experience

Complete in detail with your advisor. Indicate dates when sections/specific work will be due to faculty for review.

PLAN FOR COMPLETION OF CULMINATING EXPERIENCE	
LIST COMPONENTS OR SECTIONS FOR COMPLETION	**DATES TO BE SUBMITTED TO FACULTY FOR REVIEW**
Literature review	April 30, 2024
Initial design	May 30, 2024
System first prototype	September 30, 2
Final Prototype	November 10, 2(
System Debugging and Refinements	November 20, 2(
Final Report	November 30, 2(

I PLAN TO COMPLETE MY DEGREE IN: ☑ **FALL** ☐ **SPRING** ☐ **SUMMER** **YEAR** 2024

I have reviewed the above with my committee members and agree to the terms of the projected timeline

Naisarg Halvadiya Digitally signed by Naisarg Halvadiya Date: 2024.03.26 13:17:02 -07'00' 03/26/2024

STUDENT SIGNATURE DATE

COMPLETE PAGE 2

Name Nalsary Dhanmendra Harvadiya Student ID 923613129

8. IF YOUR PROJECT INVOLVES RESEARCH WITH HUMANS, ANIMALS OR BIOLOGICAL SPECIMENS (cells, tissues, etc.), YOU MUST PROVIDE ADDITIONAL DOCUMENTATION:

Depending on the nature of your project, SF State may require you to submit information to the Human and Animal Protections (HAP) unit for project **registration** or **approval**. Contact HAP at protocol@sfsu.edu or 415-338-1093.

1. **If you have received confirmation of *approval* or *determination* through HAP**: **Attach** the official approval or determination notice and select one of the following:
   - ☐ My protocol has been approved by the IRB or IACUC and I have attached the approval notice.
   - ☐ My project has been determined to be Exempt or Excepted by HAP and I have attached the notice of determination.
2. **If your project has been submitted to HAP and is under review**, provide the date submitted to HAP: _____, and indicate the submission type: ☐ Application for Determination of Exemption ☐ Protocol ☐ Email inquiry
3. **If your research is covered under someone else's protocol**, you must register with HAP, and indicate one of the following:
   - ☐ My project is covered under an SFSU faculty member's protocol. Faculty name:_____ Protocol #: _____
   - ☐ My project is covered under an approved protocol at another institution. Provide the following information:
     Faculty name: _____ Institution name:_____ Protocol #:_____
4. **If you are UNSURE if your project is considered human subjects research**, submit an Application for Determination of Exemption to HAP: http://research.sfsu.edu/protocol//
5. **If you are UNSURE if your project is considered animal research**, contact HAP at protocol@sfsu.edu or 415-338-1093.
6. **If you are conducting research using biological specimens** (e.g., tissues, cells, etc.), you must register the work with HAP and attach their official notice of exemption or approval.

IMPORTANT: You may not begin your research until you receive an approval or exemption notice from HAP.

Do not complete this section if you are submitting your PCE through DocuSign

9. Supervising committee: must include a minimum of two TENURED or TENURE TRACK faculty members from the student's major department

FOR COMMITTEE CHAIR:

I will be available to work with students (1) during winter break ☐ YES ☐ NO (2) during the summer months ☐ YES ☐ NO

Committee Chair:

@sfsu.edu

SIGNATURE | *Clearly* TYPE/PRINT **First and Last NAME**, RANK AND DEPT. | EMAIL (REQUIRED)

Other committee member(s):

2nd _____ SIGNATURE _____ *Clearly* TYPE/PRINT **First and Last NAME**, RANK AND DEPT.

3rd _____ SIGNATURE _____ *Clearly* TYPE/PRINT **First and Last NAME**, RANK AND DEPT.

10. Department chair/graduate coordinator: I have reviewed the above proposal including the composition of the supervising committee and find it acceptable for meeting the culminating experience requirement for the master's degree in the major indicated.

SIGNATURE | TYPE/ PRINT **First and Last NAME** AND TITLE | DATE

FOR GRADUATE STUDIES OFFICE USE ONLY

ORSP determination: Approved protocol # _____ Exempt # _____ Not Human/Animal Subjects or Research ☐

Date of ORSP determination (if known): _____

Accepted by Division of Graduate Studies_____ Date _____

DocuSign

Certificate Of Completion

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Naisarg Dharmendra Halvadiya nhalvadiya@sfsu.edu Security Level: DocuSign.email ID: 1 3/26/2024 1:27:52 PM **Electronic Record and Signature Disclosure:** Accepted: 3/22/2024 5:23:40 PM ID: a5f1dd6b-4c62-4120-aa97-a84e739320c1	**Completed** Using IP Address: 130.212.92.188	Sent: 3/26/2024 1:27:50 PM Viewed: 3/26/2024 1:28:05 PM Signed: 3/26/2024 1:32:23 PM
Shahrukh Humayoun humayoun@sfsu.edu Co-PI San Francisco State University Security Level: Email, Account Authentication (Optional) **Electronic Record and Signature Disclosure:** Not Offered via DocuSign	DocuSigned by: Shahrukh Humayoun 94E06432EAE246C... Signature Adoption: Pre-selected Style Using IP Address: 73.170.138.93	Sent: 3/26/2024 1:32:24 PM Viewed: 3/26/2024 2:36:43 PM Signed: 3/26/2024 2:39:57 PM
Jingyi Wang jingyiwang@sfsu.edu Assistant Professor San Francisco State University Security Level: Email, Account Authentication (Optional) **Electronic Record and Signature Disclosure:** Not Offered via DocuSign	DocuSigned by: Jingyi Wang 7CFEB96626BA4CF... Signature Adoption: Pre-selected Style Using IP Address: 70.138.194.171 Signed using mobile	Sent: 3/26/2024 2:39:59 PM Viewed: 3/26/2024 9:09:17 PM Signed: 3/26/2024 9:09:57 PM
Dr. Hui Yang huiyang@sfsu.edu San Francisco State University Security Level: Email, Account Authentication (Optional) **Electronic Record and Signature Disclosure:** Not Offered via DocuSign	DocuSigned by: Yang Hui 7512C57EEF6A46F... Signature Adoption: Drawn on Device Using IP Address: 24.130.111.174	Sent: 3/26/2024 9:09:58 PM Viewed: 4/1/2024 6:05:43 PM Signed: 4/1/2024 6:06:50 PM

Signer Events	Signature	Timestamp
Mary Menees mmenees@sfsu.edu Business Officer and Culminating Experience Coordinator San Francisco State University Security Level: Email, Account Authentication (Optional) **Electronic Record and Signature Disclosure:** Accepted: 11/9/2018 2:39:40 PM ID: 60c95341-888a-437e-8033-bd0b7e6f1899	DocuSigned by: 3D68EB20C83143E... Signature Adoption: Uploaded Signature Image Using IP Address: 130.212.186.246	Sent: 4/1/2024 6:06:52 PM Viewed: 4/3/2024 10:30:54 AM Signed: 4/3/2024 10:31:21 AM

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Carbon Copy Events	Status	Timestamp
Naisarg Dharmendra Halvadiya nhalvadiya@sfsu.edu Security Level: Email, Account Authentication (Optional) **Electronic Record and Signature Disclosure:** Accepted: 2/22/2024 11:31:12 AM ID: 9a3bb003-c568-4835-a518-3ed134c2dfd4	COPIED	Sent: 4/3/2024 10:31:23 AM Viewed: 4/3/2024 11:07:18 AM

Witness Events	Signature	Timestamp

Notary Events	Signature	Timestamp

Envelope Summary Events	Status	Timestamps
Envelope Sent	Hashed/Encrypted	3/26/2024 1:27:50 PM
Certified Delivered	Security Checked	4/3/2024 10:30:54 AM
Signing Complete	Security Checked	4/3/2024 10:31:21 AM
Completed	Security Checked	4/3/2024 10:31:23 AM

Payment Events	Status	Timestamps

Electronic Record and Signature Disclosure

ELECTRONIC RECORD AND SIGNATURE DISCLOSURE

From time to time, San Francisco State University (we, us or Company) may be required by law to provide to you certain written notices or disclosures. Described below are the terms and conditions for providing to you such notices and disclosures electronically through your DocuSign, Inc. (DocuSign) Express user account. Please read the information below carefully and thoroughly, and if you can access this information electronically to your satisfaction and agree to these terms and conditions, please confirm your agreement by clicking the 'I agree' button at the bottom of this document.

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If you decide to receive notices and disclosures from us electronically, you may at any time change your mind and tell us that thereafter you want to receive required notices and disclosures only in paper format. How you must inform us of your decision to receive future notices and disclosure in paper format and withdraw your consent to receive notices and disclosures electronically is described below.

Consequences of changing your mind

If you elect to receive required notices and disclosures only in paper format, it will slow the speed at which we can complete certain steps in transactions with you and delivering services to you because we will need first to send the required notices or disclosures to you in paper format, and then wait until we receive back from you your acknowledgment of your receipt of such paper notices or disclosures. To indicate to us that you are changing your mind, you must withdraw your consent using the DocuSign 'Withdraw Consent' form on the signing page of your DocuSign account. This will indicate to us that you have withdrawn your consent to receive required notices and disclosures electronically from us and you will no longer be able to use your DocuSign Express user account to receive required notices and consents electronically from us or to sign electronically documents from us.

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Unless you tell us otherwise in accordance with the procedures described herein, we will provide electronically to you through your DocuSign user account all required notices, disclosures, authorizations, acknowledgements, and other documents that are required to be provided or made available to you during the course of our relationship with you. To reduce the chance of you inadvertently not receiving any notice or disclosure, we prefer to provide all of the required notices and disclosures to you by the same method and to the same address that you have given us. Thus, you can receive all the disclosures and notices electronically or in paper format through the paper mail delivery system. If you do not agree with this process, please let us know as described below. Please also see the paragraph immediately above that describes the consequences of your electing not to receive delivery of the notices and disclosures electronically from us.

How to contact San Francisco State University:

You may contact us to let us know of your changes as to how we may contact you electronically, to request paper copies of certain information from us, and to withdraw your prior consent to receive notices and disclosures electronically as follows:
To contact us by email send messages to: jpleung@sfsu.edu

To advise San Francisco State University of your new e-mail address

To let us know of a change in your e-mail address where we should send notices and disclosures electronically to you, you must send an email message to us at jpleung@sfsu.edu and in the body of such request you must state: your previous e-mail address, your new e-mail address. We do not require any other information from you to change your email address..

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To inform us that you no longer want to receive future notices and disclosures in electronic format you may:

> i. decline to sign a document from within your DocuSign account, and on the subsequent page, select the check-box indicating you wish to withdraw your consent, or you may;
>
> ii. send us an e-mail to jpleung@sfsu.edu and in the body of such request you must state your e-mail, full name, IS Postal Address, telephone number, and account number. We do not need any other information from you to withdraw consent.. The consequences of your withdrawing consent for online documents will be that transactions may take a longer time to process..

Required hardware and software

Operating Systems:	Windows2000? or WindowsXP?
Browsers (for SENDERS):	Internet Explorer 6.0? or above
Browsers (for SIGNERS):	Internet Explorer 6.0?, Mozilla FireFox 1.0, NetScape 7.2 (or above)
Email:	Access to a valid email account
Screen Resolution:	800 x 600 minimum
Enabled Security Settings:	• Allow per session cookies

	• Users accessing the internet behind a Proxy Server must enable HTTP 1.1 settings via proxy connection

** These minimum requirements are subject to change. If these requirements change, we will provide you with an email message at the email address we have on file for you at that time providing you with the revised hardware and software requirements, at which time you will have the right to withdraw your consent.

Acknowledging your access and consent to receive materials electronically

To confirm to us that you can access this information electronically, which will be similar to other electronic notices and disclosures that we will provide to you, please verify that you were able to read this electronic disclosure and that you also were able to print on paper or electronically save this page for your future reference and access or that you were able to e-mail this disclosure and consent to an address where you will be able to print on paper or save it for your future reference and access. Further, if you consent to receiving notices and disclosures exclusively in electronic format on the terms and conditions described above, please let us know by clicking the 'I agree' button below.

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- I can print on paper the disclosure or save or send the disclosure to a place where I can print it, for future reference and access; and
- Until or unless I notify San Francisco State University as described above, I consent to receive from exclusively through electronic means all notices, disclosures, authorizations, acknowledgements, and other documents that are required to be provided or made available to me by San Francisco State University during the course of my relationship with you.